Fake image from ICLight (relighting):
identity preserved, lighting synthesised

Legend (Icons)

- ✅ Correct prediction (FAKE)
- ⚠️ Partially correct / Uncertain
- ❌ Wrong prediction (REAL)

1) FakeShield (22B) — ❌ REAL

1. Whether the picture has been tampered with / Description of the tampered area:
The picture has not been tampered with. It appears to be a portrait photograph captured under strong uplight, but it is otherwise a real, unaltered image.

2. Judgment basis:

- The image shows a single subject against a dim background, lit predominantly from below.
- The skin texture is detailed and includes pores, stubble, and individual hair strands.
- There are no compositing edges, jagged pixel transitions, or mismatched resolutions.
- Colour temperature is consistent across the face, hair, and background.
- Specular highlights on the eyes correspond to the apparent light source.
- Physical proportions are correct and there are no contradictory shadows.
- All these details support the conclusion that this is a real, unaltered photograph.

2) SIDA-13B-description — ⚠️ PARTIAL (leans REAL)

Type: borderline
Areas: **<suspect region>** Subtle light/shadow inconsistencies along the upper temples and the nose-cheek transition, where the apparent light source from below does not produce the expected fall-off pattern on certain micro-features.
Tampered Content: Possibly relighting / illumination editing.
Visual Inconsistencies:
<lighting> The terminator line between lit and shadowed skin is slightly softer than what natural uplight produces.
<shadows> Cast shadows from the nose and chin are weaker than expected.
<texture> Skin micro-texture is preserved everywhere — no smoothing artefacts.
<boundary> No compositing seams detected.
Summary: The image shows weak signals consistent with localised light / shadow editing, but the underlying face content appears authentic. **Verdict:** borderline — leaning real.

3) FFAA (7B) — ❌ REAL

Image description: A close-up portrait of a person under dramatic uplight. Skin texture is rich with visible pores and stubble, hair detail is preserved at single-strand level, and the eyes show sharp catchlights. The background is dim and uniform.
Forgery reasoning: The skin micro-texture, hair detail, and eye reflections all appear authentic. There is no evidence of generative skin smoothing or facial re-synthesis. The dramatic lighting is unusual but does not by itself indicate manipulation; theatrical portraits frequently use strong uplight.
Analysis result: real, **Forgery type:** None
Match score: 0.8634; **Difficulty:** hard

4) M2F2-Det (7B) — ⚠️ PARTIAL

<s> The image looks largely natural, with consistent skin texture and hair detail. The dramatic uplight is unusual; this could indicate post-processing or simply a creative portrait setup. Likely real with low confidence. </s>

5) FakeVLM (7B) — ❌ REAL

This image appears to be a real photograph. The skin micro-texture is consistent and there are no obvious artefacts of generative synthesis.

6) Veritas (8B, prior SOTA) — ❌ REAL

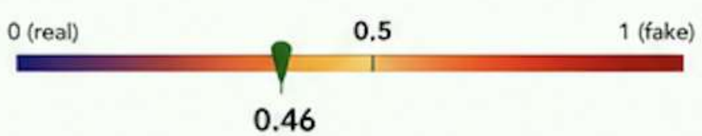
<fast> The image shows a portrait under strong uplight, but the face itself looks genuine. **</fast>**
<planning> Verify (1) skin micro-texture vs. AI smoothing, (2) eye-reflection consistency with the apparent light source, (3) hair-edge integrity, and (4) cohesion between facial expression and lighting. **</planning>**
<reasoning> The skin shows pores and stubble, rarely preserved by fully synthesised faces. The eyes have sharp catchlights consistent with a strong source from below. Hair contour is natural and the rim-light is physically plausible. The shadow pattern is unusual but consistent within itself. I do not see strong cues of synthesis. **</reasoning>**
<reflection> Could the image be relit rather than synthesised? In principle yes, but I see no obvious sign of digital re-illumination — the chiaroscuro looks photographically captured. **</reflection>**
<conclusion> The face content is authentic and the lighting, while dramatic, is internally coherent. **</conclusion>**
<answer> real </answer>

7) GenStack (Ours, dual-branch + MoE) — ✅ FAKE

Branch 1 — PACT (discriminative, frozen CLIP + 8 FPT + 128 prototypes)

$p_v(x) = 0.46$ (continuous, just below the 0.5 threshold — uncertain)

Heatmap concentration: cheek-shadow boundary on the upper-left of the face.



Interpretation: Near the decision boundary → uncertain.

Branch 2 — Prism-SFT (multi-view MLLM: RGB + FFT + noise residual)

<plan> Inspect lighting coherence, frequency signature, and noise residual. </plan>
<examine region="lighting_coherence"> Multi-source incoherence between face, hair, and background suggests per-region re-rendering. </examine>
<examine region="frequency"> Faint cross-shaped energy concentration in FFT — a known fingerprint of diffusion U-Net samplers. </examine>
<examine region="noise_residual"> Low-amplitude smoothly-varying residual follows the skin contour — consistent with relighting (re-rendering existing pixels) rather than splicing. </examine>
<conclude> Three independent channels jointly indicate a diffusion-relit face. Identity is real, illumination is synthesised. </conclude>
<answer> *fake* </answer>

$b_p(x) = 1$

Per-Generator MoE Fusion (Key Innovation)

Routing (oracle gate): $g(x) = \text{ICLight}$ → routed to expert f_{ICLight}

Input to expert: $(p_v, b_p) = (0.46, 1)$

Expert f_{ICLight} is specialized on ICLight relighting artifacts (illumination field + noise statistics).

Final probability: $\hat{y}(x) = 0.91$ → threshold 0.5 → **FAKE** ✅

On ICLight, f_{ICLight} has learned a non-monotone rule that "PACT-uncertainty + Prism-fake = strong fake signal" — a pattern no global combiner can encode.